# Daily Practice 2

## Section 1: Multiple Choice Questions

*No calculator is allowed for these questions.*

**1.** What is the value of $\lim_{x \to 3} \frac{x^2-x-6}{x^2-9}$ ?

(A) $0$
(B) $\frac{5}{6}$
(C) $1$
(D) The limit does not exist.

**2.** What is the value of $\lim_{x \to 0} \frac{\sqrt{x+16}-4}{x}$ ?

(A) $0$
(B) $\frac{1}{8}$
(C) $\frac{1}{4}$
(D) The limit does not exist.

**3.** What is the value of $\lim_{x \to 0} \frac{\sin(4x)}{5x}$ ?

(A) $0$
(B) $\frac{4}{5}$
(C) $\frac{5}{4}$
(D) $1$

**4.** The function $f$ is defined by $f(x) = \frac{|x-2|}{x-2}$ for all $x \neq 2$. What is the value of $\lim_{x \to 2^-} f(x)$ ?

(A) $-1$
(B) $0$
(C) $1$
(D) The limit does not exist.

**5.** Which of the following lines is a vertical asymptote to the graph of $f(x) = \frac{x^2-1}{x^2+2x-3}$ ?

(A) $x = -3$ only
(B) $x = 1$ only
(C) $x = -3$ and $x = 1$
(D) $y = 1$

**6.** Which of the following lines is a horizontal asymptote to the graph of $y = \frac{2x^3-5x+1}{7x^3+4x^2}$ ?

(A) $y = 0$
(B) $y = \frac{2}{7}$

(C) $y = \frac{1}{4}$
(D) There is no horizontal asymptote.

**7.** What is the value of $\lim_{x \to -\infty} \frac{5x}{\sqrt{9x^2+1}}$ ?

(A) $-\frac{5}{3}$
(B) $0$
(C) $\frac{5}{3}$
(D) The limit does not exist.

**8.** Let $f$ be the piecewise function defined below:

$$f(x) = \begin{cases} 2x^2 - 1, & \text{for } x < 2 \\ 3x + 1, & \text{for } x \geq 2 \end{cases}$$

What is the value of $\lim_{x \to 2} f(x)$ ?

(A) $7$
(B) $8$
(C) $9$
(D) The limit does not exist.

**9.** Let $g(x) = \begin{cases} kx^2 - 3, & \text{for } x < 3 \\ 2x + k, & \text{for } x \geq 3 \end{cases}$. For what value of the constant $k$ is $g$ continuous at $x = 3$ ?

(A) $1$
(B) $\frac{9}{8}$
(C) $\frac{9}{10}$
(D) $\frac{10}{9}$

**10.** Let $f$ be a continuous function on the closed interval $[-2, 4]$. If $f(-2) = 5$ and $f(4) = -1$, which of the following MUST be true?

(A) $f(x) = 0$ for some value $x$ between $-2$ and $4$.
(B) $f(0) = 2$.
(C) $f'(x) < 0$ for all $x$ between $-2$ and $4$.
(D) The maximum value of $f(x)$ on the interval is $5$.

**11.** What is the value of $\lim_{x \to 0} \frac{1-\cos(x)}{2x}$ ?

(A) $0$
(B) $\frac{1}{2}$
(C) $1$
(D) The limit does not exist.

**12.** The function $h(x)$ has a removable discontinuity at $x = -4$. Which of the following statements must be true?

(A) $h(-4)$ is undefined.
(B) $\lim_{x \to -4} h(x)$ does not exist.
(C) $\lim_{x \to -4} h(x)$ exists, but it does not equal $h(-4)$.
(D) The line $x = -4$ is a vertical asymptote of the graph of $h$.

**13.** $\lim_{x \to \infty} \frac{\sin x}{x}$ is

(A) $0$
(B) $1$
(C) $\infty$
(D) nonexistent

**14.** For which of the following functions does $\lim_{x \to 3} f(x) = \infty$ ?

(A) $f(x) = \frac{x^2-9}{x-3}$
(B) $f(x) = \frac{x-3}{x^2-9}$
(C) $f(x) = \frac{1}{(x-3)^2}$
(D) $f(x) = \frac{1}{\sqrt{x^2+9}}$

**15.** Let $f(x) = \frac{x^2-4x+3}{x-1}$. If $f$ is extended to a continuous function $g$ defined for all real numbers, what is the value of $g(1)$ ?

(A) $-2$
(B) $0$
(C) $2$
(D) $4$

# Section 2: Free Response Questions

_Show all your work.

**FRQ 1.** Let $f$ be the piecewise function defined by:

$$f(x) = \begin{cases} \frac{x^2-25}{x-5}, & \text{for } x < 5 \\ k, & \text{for } x = 5 \\ \sqrt{x+4} + m, & \text{for } x > 5 \end{cases}$$

where $k$ and $m$ are constants.

**(a)** Evaluate $\lim_{x \to 5^-} f(x)$. Use proper limit notation in your work.

**(b)** If $f$ is continuous at $x = 5$, what must be the value of $k$? Justify your answer using the definition of continuity.

**(c)** Find the value of $m$ so that $f$ is continuous at $x = 5$. Show the limit setup that leads to your equation.

**FRQ 2.** A particle moves along a straight line so that its velocity $v(t)$, in meters per second, is modeled by a continuous function for $0 \le t \le 12$. Selected values of $v(t)$ are given in the table below.

| $t$ **(seconds)** | $0$ | $3$ | $5$ | $8$ | $12$ |
|---|---|---|---|---|---|
| $v(t)$ (m/s) | $4$ | $1$ | $-2$ | $5$ | $9$ |

**(a)** Is there a time $t$, for $3 \le t \le 8$, such that the particle is at rest (meaning its velocity is exactly $0$ m/s)? Justify your answer using a relevant theorem and the values from the table.

**FRQ 3.** Consider the rational function $g(x) = \frac{2x^2+6x}{x^2-3x-18}$.

**(a)** Find $\lim_{x \to -3} g(x)$. Show the algebraic steps that lead to your answer.

**(b)** Write an equation for each vertical asymptote to the graph of $g(x)$. Justify why it is a vertical asymptote and not a removable discontinuity (hole).

**(c)** Write an equation for the horizontal asymptote to the graph of $g(x)$. Show the appropriate limit expression that justifies your answer.